



The Derived Calabi–Yau Auslander–Iyama correspondence

Homological Algebra in Algebraic Geometry and Representation Theory

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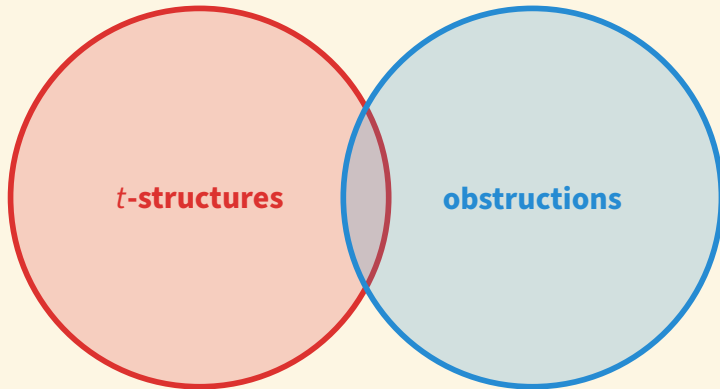
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Uniqueness of enhancements of triangulated categories



**Lunts, Orlov, Canonaco,
Stellari, Neeman**

Jasso, Muro

The derived Auslander correspondence

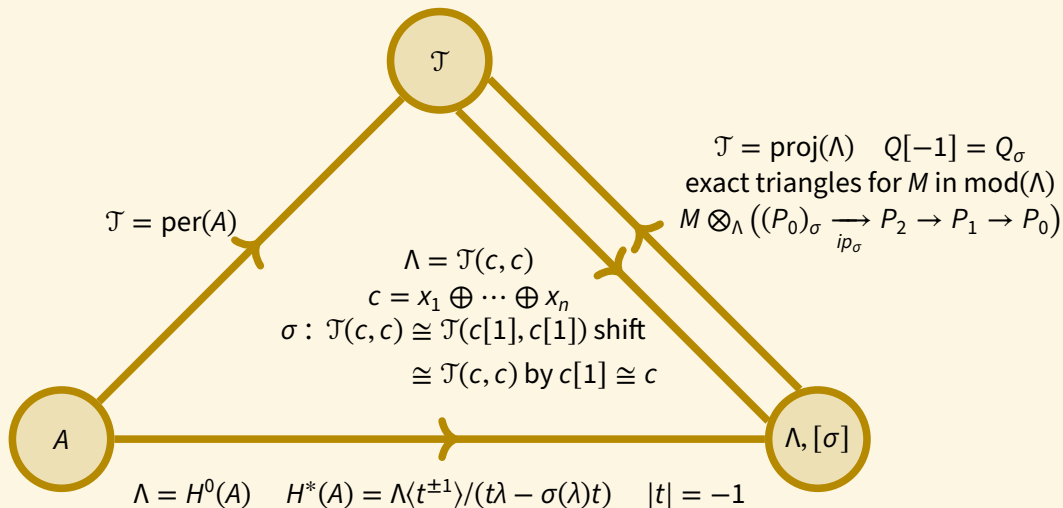
Theorem (Muro 2022)

Over a perfect field k , there is a bijection between equivalence classes of:

1. A a **differential graded algebra (DGA)** such that:
 - a. $H^0(A)$ is a finite-dimensional basic algebra,
 - b. $A \in \text{per}(A)$ is an additive generator.
2. $\mathcal{T}$ an idempotent-complete, hom-finite **triangulated category** with finitely-many indecomposables $x_1, \dots, x_n$ up to isomorphism.
3. $(\Lambda, [\sigma])$ where:
 - a. Λ is a **self-injective basic algebra**,
 - b. $[\sigma] \in \text{Out}(\Lambda)$ such that Λ is **σ -twisted 3-periodic**, i.e. there is an exact sequence of Λ -bimodules

$$\Lambda_\sigma \xhookrightarrow{i} \underbrace{P_2 \rightarrow P_1 \rightarrow P_0}_{\text{projective-injective}} \xrightarrow{p} \Lambda.$$

The derived Auslander correspondence

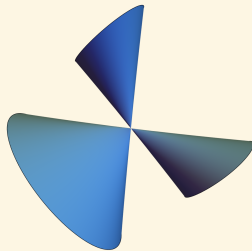


The derived Auslander correspondence

Example

The following $\mathcal{T}$ fit in the derived Auslander correspondence:

1. $\mathcal{T} = \underline{\text{mod}}(B)$ for B a finite-dimensional self-injective representation finite algebra.
2. $\mathcal{T} = D^{sg}(R) = D^b(R)/\text{per}(R)$ for $\text{Spec}(R)$ a complete simple surface singularity.



Cluster tilting

If $\mathcal{T}$ is an idempotent-complete, hom-finite triangulated category, an object c in $\mathcal{T}$ is **$d\mathbb{Z}$ -cluster tilting** for some $d \geq 1$ if:

$$\begin{aligned}c[d] &\cong c, \\ \text{add}(c) &= \{x \in \mathcal{T} \mid \mathcal{T}(x, c[i]) = 0 \ \forall 0 < i < d\} \\ &= \{x \in \mathcal{T} \mid \mathcal{T}(c, x[i]) = 0 \ \forall 0 < i < d\}.\end{aligned}$$

For $d = 1$, an object is $1\mathbb{Z}$ -cluster tilting $\Leftrightarrow$ it is an additive generator.

The derived Auslander–Iyama correspondence

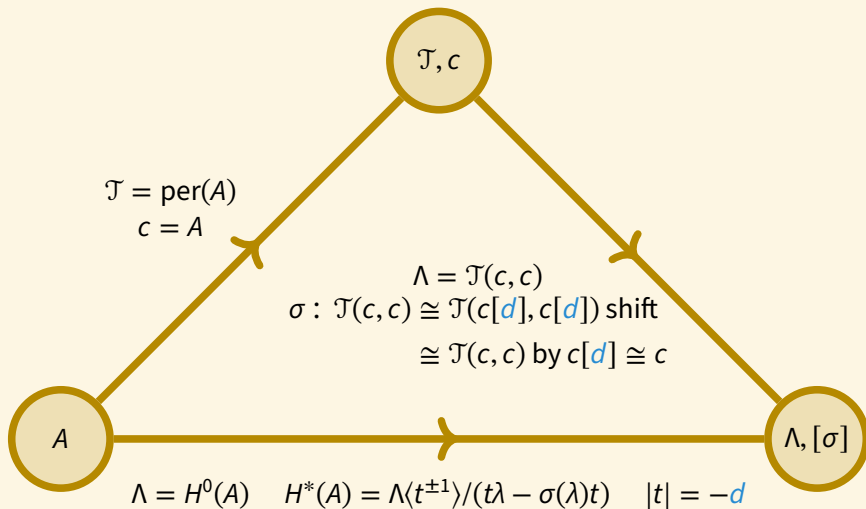
Theorem (Jasso and Muro 2023)

Over a perfect field k , for $d \geq 1$, there is a bijection between equivalence classes of:

1. A a **differential graded algebra** such that:
 - a. $H^0(A)$ is a finite-dimensional basic algebra,
 - b. $A \in \text{per}(A)$ is $d\mathbb{Z}$ -cluster tilting.
2. $(\mathcal{T}, c)$ where:
 - a. $\mathcal{T}$ is an idempotent-complete, hom-finite, **algebraic triangulated category**,
 - b. c is a basic $d\mathbb{Z}$ -cluster tilting object in $\mathcal{T}$.
3. $(\Lambda, [\sigma])$ where:
 - a. Λ is a **self-injective basic algebra**,
 - b. $[\sigma] \in \text{Out}(\Lambda)$ such that Λ is σ -**twisted $(d+2)$ -periodic**, i.e. there is an exact sequence of Λ -bimodules

$$\Lambda_\sigma \hookrightarrow \underbrace{P_{d+1} \rightarrow \cdots \rightarrow P_0}_{\text{projective-injective}} \twoheadrightarrow \Lambda.$$

The derived Auslander–Iyama correspondence

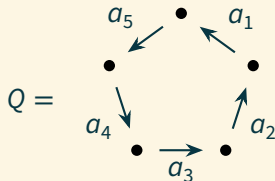


The derived Auslander–Iyama correspondence

Example

The following $(\mathcal{T}, c)$ fit in the derived Auslander–Iyama correspondence:

1. $\mathcal{T} = D^{sg}(R)$ for $\text{Spec}(R)$ a complete cDV singularity with a crepant resolution ($d = 2$), Donovan and Wemyss 2016, Keller's appendix in Jasso and Muro 2023.
2. $\mathcal{T} = \mathcal{C}(\Gamma) = \text{per}(\Gamma)/\text{fd}(\Gamma)$, $c = [\Gamma]$, the Amiot–Guo–Keller cluster category of the following DGAs:
 - a. $\Gamma = \Pi_{d+1}(B)$ the derived preprojective algebra of a d -representation finite algebra B . Here $\Lambda = \Pi_{d+1}(B)$ is the ordinary preprojective algebra.
 - b. $\Gamma = \Gamma(Q, W)$ the Ginzburg DGA of a quiver Q with potential W with finite-dimensional self-injective Jacobian algebra $\Lambda = J(Q, W)$ ($d = 2$), Amiot 2009, Keller and Liu 2024.



$$W = a_1 \cdots a_5$$

The bimodule Calabi–Yau property

Let A be a DGA. The **dual** $DM = \text{hom}(M, k)$ of an A -bimodule M is an A -bimodule with

$$(a \cdot f \cdot b)(m) = (-1)^{|a|(|f|+|b|+|m|)} f(b \cdot m \cdot a).$$

The DGA A is **bimodule n -Calabi–Yau** if there is a quasi-isomorphism of A -bimodules

$$A[n] \xrightarrow{\sim} DA.$$

The DGAs associated to the previous examples via the derived Auslander–Iyama correspondence are bimodule d -Calabi–Yau.

Question

How does the bimodule n -Calabi–Yau property fit into the derived Auslander–Iyama correspondence?

Calabi–Yau triangulated categories and the Nakayama automorphism

A triangulated category $\mathcal{T}$ is ***n-Calabi–Yau*** if the n -fold shift functor is a graded Serre functor, i.e. there are natural isomorphisms compatible with the shift

$$\mathcal{T}(x, y[n]) \cong D\mathcal{T}(y, x).$$

If A is a bimodule n -Calabi–Yau DGA $\Rightarrow \text{per}(A)$ is n -Calabi–Yau.

Given a pair $(\mathcal{T}, c)$ where:

- $\mathcal{T}$ is idempotent-complete d -Calabi–Yau,
- c is a basic $d\mathbb{Z}$ -cluster tilting object in $\mathcal{T}$,

for some $d \geq 1$, then $\Lambda = \mathcal{T}(c, c)$ is ν -twisted $(d + 2)$ -periodic for $\nu \in \text{Aut}(\Lambda)$ the ***Nakayama automorphism***,

$${}_{\nu}\Lambda \cong D\Lambda.$$

Hochschild cohomology

Given a graded algebra A , the **Hochschild cohomology**

$$\mathrm{HH}^{\star,*}(A) = \mathrm{Ext}_{A^e}^{\star,*}(A, A)$$

is a **Gerstenhaber algebra**, i.e.

- a graded **commutative** algebra,
- a graded **Lie** algebra shifted by -1 ,

and both structures are compatible in the following way

$$[x, y \cdot z] = [x, y] \cdot z + (-1)^{(|x|-1)|y|} y \cdot [x, z].$$

Twisted Laurent polynomial algebras and Massey products

Let Λ be a σ -twisted $(d + 2)$ -periodic self-injective basic algebra,

$$\eta : \Lambda_\sigma \hookrightarrow \underbrace{P_{d+1} \rightarrow \cdots \rightarrow P_0}_{\text{projective-injective}} \twoheadrightarrow \Lambda,$$

The **d -sparse σ -twisted Laurent polynomial algebra** is

$$\Lambda(\sigma, d) = \frac{\Lambda\langle t^{\pm 1} \rangle}{(t\lambda - \sigma(\lambda)t)}, \quad |\Lambda| = 0, \quad |t| = -d.$$

The inclusion of the degree 0 part $\Lambda \subset \Lambda(\sigma, d)$ induces a map such that

$$\begin{aligned} \mathrm{HH}^{d+2, -d}(\Lambda(\sigma, d)) &\longrightarrow \mathrm{HH}^{d+2}(\Lambda, \Lambda_\sigma) = \mathrm{Ext}_{\Lambda^e}^{d+2}(\Lambda, \Lambda_\sigma), \\ \exists! m &\longmapsto [\eta], \end{aligned}$$

with $\frac{1}{2}[m, m] = 0$. We call m a **universal Massey product (UMP)** of $(\Lambda, [\sigma])$.

The derived Calabi–Yau Auslander–Iyama correspondence

If Λ is a self-injective basic algebra with **Nakayama automorphism** $\nu \in \text{Aut}(\Lambda)$ then $\text{HH}^{\star,*}(\Lambda(\nu, d))$ is equipped with a degree -1 **Batalin–Vilkovisky operator** Δ such that

$$[x, y] = \Delta(x \cdot y) - \Delta(x) \cdot y - (-1)^{|x|} x \cdot \Delta(y).$$

Theorem (Jasso and Muro 2025b)

Over a perfect field k , for $d \geq 1$, there is a bijection between equivalence classes of:

1. A a **differential graded algebra** such that:
 - a. $A \in \text{per}(A)$ is basic $d\mathbb{Z}$ -cluster tilting,
 - b. A is bimodule d -Calabi–Yau.
2. Λ a **self-injective basic algebra** such that:
 - a. Λ is ν -twisted $(d+2)$ -periodic,
 - b. $(\Lambda, [\nu])$ has a universal Massey product $m \in \text{HH}^{d+2,-d}(\Lambda(\nu, d))$ such that

$$\Delta(m) = 0.$$

More Massey products

A DGA $\rightsquigarrow (H^*A, m_3, m_4, \dots)$ A_∞ -algebra **minimal model**.

$$m_3(a_1, a_2, a_3) \in \langle a_1, a_2, a_3 \rangle, \quad a_i \in H^*A, \quad \textbf{Massey product.}$$

The **universal Massey product**¹ (**UMP**) of a DGA A with d -sparse cohomology

$$m_A = \{m_{d+2}\} \in \mathrm{HH}^{d+2, -d}(H^*A), \quad \frac{1}{2}[m_A, m_A] = 0.$$

Under the derived Auslander–Lyons correspondence, the following notions agree:

- The universal Massey product $m_A \in \mathrm{HH}^{d+2, -d}(H^*A)$ of A .
- The universal Massey product $m_{\Lambda, [\sigma]} \in \mathrm{HH}^{d+2, -d}(\Lambda(\sigma, d))$ of $(\Lambda, [\sigma])$.

¹Baues and Dreckmann 1989; Benson, Krause, and Schwede 2004; Kaledin 2007...

Formality-like results for algebras

Theorem (Kadeishvili 1988)

If A is a DGA with $\mathrm{HH}^{n+2,-n}(H^*A) = 0$, $n > 0$, then any other DGA B with $H^*B = H^*A$ is $B \simeq A$.

A **d -Massey algebra** (A, m) is a d -sparse graded algebra A and

$$m \in \mathrm{HH}^{d+2,-d}(A), \quad \frac{1}{2}[m, m] = 0.$$

Its **Hochschild cohomology** $\mathrm{HH}^{\star,*}(A, m)$ is the cohomology of

$$(\mathrm{HH}^{\star,*}(A), d = [m, -]).$$

Theorem (Jasso and Muro 2023)

If A is a DGA with d -sparse cohomology and $\mathrm{HH}^{n+2,-n}(H^*A, m_A) = 0$, $n > d$, then any other DGA B with d -sparse cohomology and $(H^*B, m_B) = (H^*A, m_A)$ is $B \simeq A$.

Bimodule Hochschild cohomology

Given a graded algebra A and an A -bimodule M , we define a **bimodule Hochschild cohomology** of this pair

$$\mathrm{HH}^{\star,*}(A|M).$$

It is a Gerstenhaber algebra. Moreover, it fits into a long exact sequence

$$\dots \rightarrow \mathrm{HH}^{p,q}(A|M) \rightarrow \mathrm{HH}^{p,q}(A) \xrightarrow{\delta} \mathrm{Ext}_{A^e}^{p,q}(M, M) \rightarrow \mathrm{HH}^{p+1,q}(A|M) \rightarrow \dots$$

where

$$\delta(x) = 1_M \cdot x - x \cdot 1_M$$

vanishes iff the $\mathrm{HH}^{\star,*}(A)$ -bimodule $\mathrm{Ext}_{A^e}^{\star,*}(M, M)$ is symmetric.

Even more Massey products

A DGA and M an A -bimodule $\rightsquigarrow (H^*M, m_{0,2}^M, m_{1,1}^M, m_{2,0}^M, \dots)$ A_∞ -bimodule over $(H^*A, m_3^A, m_4^A, \dots)$ **minimal model**.

$$m_{1,1}^M(a, m, b) \in \langle a, m, b \rangle, \quad m \in H^*M, \quad a, b \in H^*A, \quad \textbf{Massey product.}$$

The **universal Massey product (UMP)** of a bimodule M over a DGA A , both with d -sparse cohomology,

$$m_M = \{m_{d+2}^A + \sum_{p+q=d+1} m_{p,q}^M\} \in \mathrm{HH}^{d+2, -d}(H^*A|H^*M), \quad \frac{1}{2}[m_M, m_M] = 0.$$

It satisfies

$$\begin{aligned} \mathrm{HH}^{d+2, -d}(H^*A|H^*M) &\longrightarrow \mathrm{HH}^{d+2, -d}(H^*A), \\ m_M &\longmapsto m_A. \end{aligned}$$

Formality-like results for bimodules

Theorem (Jasso and Muro 2025a)

If A is a DGA and M is an A -bimodule with $\mathrm{Ext}_{H^*(A)^e}^{n+1, -n}(H^*M, H^*M) = 0, n > 0$, then any other A -bimodule N with $H^*N = H^*M$ is $N \simeq M$.

A **d -Massey bimodule** (M, m_M) over a d -Massey algebra (A, m_A) consists of a d -sparse A -bimodule M and a bimodule Hochschild cohomology class with

$$\begin{aligned} \mathrm{HH}^{d+2, -d}(A|M) &\longrightarrow \mathrm{HH}^{d+2, -d}(A), \\ m_M &\longmapsto m_A, \end{aligned}$$

and $\frac{1}{2}[m_M, m_M] = 0$. Its **Hochschild cohomology** $\mathrm{HH}^{\star, *}(M, m_M)$ is the cohomology of

$$(\mathrm{Ext}_{A^e}^{\star, *}(M, M), d = [m_M, -]).$$

Formality-like results for bimodules

Theorem (Jasso and Muro 2025a)

If M is a bimodule over a DGA A , both with d -sparse cohomology, such that the $\mathrm{HH}^{\star,*}(H^*A)$ -module $\mathrm{Ext}_{H^*(A)^e}^{\star,*}(H^*M, H^*M)$ is symmetric and $\mathrm{HH}^{n+1,-n}(H^*M, m_M) = 0$ for $n > d$, then any other A -bimodule N with $(H^*N, m_N) = (H^*M, m_M)$ is $N \simeq M$.

If A is a graded algebra and $M = A$, the bimodule Hochschild cohomology is

$$\mathrm{HH}^{\star,*}(A|A) = \mathrm{HH}^{\star,*}(A)[\varepsilon]/(\varepsilon^2) = \mathrm{HH}^{\star,*}(A) \oplus \mathrm{HH}^{\star,*}(A) \cdot \varepsilon, \quad |\varepsilon| = 1.$$

If A is a DGA with d -sparse cohomology, the UMP of $M = A$ is

$$m_A + 0 \cdot \varepsilon \in \mathrm{HH}^{\star,*}(H^*A)[\varepsilon]/(\varepsilon^2).$$

Moreover, if H^*A is d -Calabi–Yau, $H^*A \cong D(H^*A)[-d]$, then the UMP of $M = DA[-d]$ is

$$m_A + \Delta(m_A) \cdot \varepsilon \in \mathrm{HH}^{\star,*}(H^*A)[\varepsilon]/(\varepsilon^2).$$

 **THE END** 

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